# Chapter 9. Discriminative Methods: Logistic Regression

Lecture 7-2 (discriminative vs generative, logit model, GLM, maximum likelihood, confounding, extensions, multinomial regression)

## Contents

**1 Discriminative Methods versus Generative Models** **2**

**2 Why Not Linear Regression?** **2**

**3 The Logistic Model** **2**

3.1 Logistic regression as a GLM . . . . . 2

**4 Estimation by Maximum Likelihood** **2**

**5 Logistic Regression on the Default Data** **3**

**6 Multiple Logistic Regression and Confounding** **4**

**7 Extensions of Logistic Regression** **4**

**8 More Than Two Classes: Multinomial Regression** **4**

# 1 Discriminative Methods versus Generative Models

| Discriminative: model $\Pr(Y = c_k \mid X = x)$ directly | Generative: model $(X, Y)$, then derive $\Pr(Y = c_k \mid X = x)$ |
|---|---|
| $K$-nearest neighbors, logistic regression, classification trees, ensembles (bagging, random forest, boosting), support vector machines | linear and quadratic discriminant analysis, naive Bayes, Gaussian mixture models |

Examples of classification: tumour cells benign/malignant; credit-card transactions legitimate/fraudulent; news stories finance/weather/entertainment/sports. We observe $(X, Y)$ with $Y$ a class label and learn $C(X)$ that predicts $Y$; often we estimate $P(Y = k \mid X)$ and classify to the most probable class.

# 2 Why Not Linear Regression?

Code $Y = 0$ (No) and $Y = 1$ (Yes) and fit $\hat{Y} = \beta_0 + \beta X$, classifying by $\hat{Y} > 0.5$? There is a model mismatch: $\mathbb{E}[Y \mid X] = P(Y = 1 \mid X)$ must lie in $[0, 1]$, but a linear model is unbounded and can give invalid probabilities; the relationship is typically nonlinear in $X$. (For two classes, the least-squares fit is in fact equivalent to LDA for the classification rule, but the probability estimates are poor.)

With more than two classes, coding e.g. $Y = 1$ (stroke), 2 (drug overdose), 3 (epileptic seizure) imposes an ordering and equal spacing that has no meaning, and different codings give different fits. Multiclass logistic regression or discriminant analysis is appropriate.

# 3 The Logistic Model

For two classes $Y \in \{c_1, c_2\}$, model $\Pr(Y = c_1 \mid X)$ through the **logit** transformation:

$$\log \frac{\Pr(Y = c_1 \mid X)}{\Pr(Y = c_2 \mid X)} = \beta_0 + \beta_1 X,$$

$$\Pr(Y = c_1 \mid X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}, \quad \Pr(Y = c_2 \mid X) = \frac{1}{1 + e^{\beta_0 + \beta_1 X}},$$

which keeps probabilities in $[0, 1]$ and gives an S-shaped curve. $\beta_1$ is the change in the **log odds** for a one-unit increase in $X$; $e^{\beta_1}$ multiplies the odds; the change in the probability itself depends on $X$, $\partial p / \partial X = \beta_1 p(1 - p)$.

## 3.1 Logistic regression as a GLM

A generalized linear model consists of a random component ($Y$ from an exponential family), a linear predictor $\eta = X^\top \beta$, and a link $g(\mathbb{E}[Y]) = \eta$. Logistic regression has $Y \sim \text{Bernoulli}(p)$ and the logit link $g(p) = \log \frac{p}{1-p}$, which is the canonical link of the Bernoulli family (the natural parameter of $p^y (1 - p)^{1-y} = \exp\{y \log \frac{p}{1-p} + \log(1 - p)\}$). Hence $\log \frac{p}{1-p} = X^\top \beta$.

# 4 Estimation by Maximum Likelihood

The likelihood of the observed zeros and ones is

$$\mathcal{L}(\beta_0, \beta) = \prod_{i: y_i = 1} p(x_i) \prod_{j: y_j = 0} \{1 - p(x_j)\}, \qquad \ell(\beta_0, \beta_1) = \sum_i \Big[ y_i (\beta_0 + \beta_1 x_i) - \log \left(1 + e^{\beta_0 + \beta_1 x_i}\right) \Big].$$

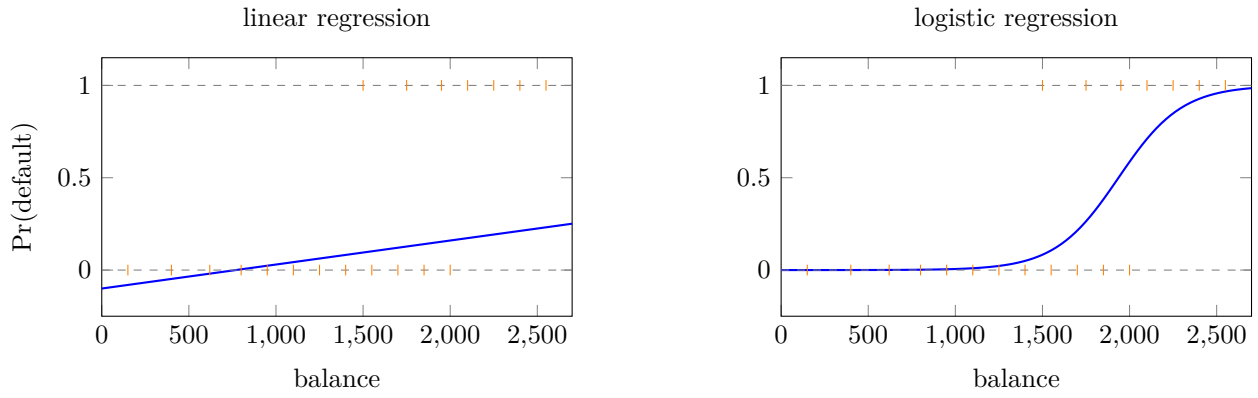



**Figure 1:** Default data (schematic points; orange ticks are $Y = 0$ at the bottom and $Y = 1$ at the top). A linear fit can give probabilities below 0 (and above 1); the fitted logistic curve $\hat{p} = e^{-10.65+0.0055X}/(1 + e^{-10.65+0.0055X})$ stays in $[0, 1]$.

**Theorem: score equations and concavity**

With $p_i = p(x_i)$ and design rows $\tilde{x}_i = (1, x_i)^\top$,

$$\nabla \ell = \sum_i (y_i - p_i)\tilde{x}_i = \tilde{X}^\top (y - p), \qquad \nabla^2 \ell = -\tilde{X}^\top W \tilde{X}, \; W = \mathrm{diag}\{p_i(1 - p_i)\}.$$

$\ell$ is concave, so the MLE solves $\tilde{X}^\top(y-p) = 0$ and is found by Newton–Raphson, equivalently **iteratively reweighted least squares**: $\beta^{\text{new}} = (\tilde{X}^\top W \tilde{X})^{-1} \tilde{X}^\top W z$ with working response $z = \tilde{X}\beta + W^{-1}(y - p)$.

**Proof.** $\frac{d}{d\eta} \log(1 + e^\eta) = \frac{e^\eta}{1+e^\eta} = p$ and $\frac{dp}{d\eta} = p(1-p)$; differentiate termwise. The Hessian is negative semidefinite. A Newton step $\beta + (\tilde{X}^\top W \tilde{X})^{-1} \tilde{X}^\top (y - p)$ rearranges into the weighted least-squares form. ■

The asymptotic covariance of $\hat{\beta}$ is $(\tilde{X}^\top \hat{W} \tilde{X})^{-1}$ (inverse Fisher information), which gives the standard errors and $z$-statistics. If the classes are perfectly separable, the likelihood increases without bound as $\|\beta\| \to \infty$ and the MLE does not exist (estimates become unstable — see also Ch. 10).

## 5 Logistic Regression on the Default Data

| | Coefficient | Std. error | $z$-statistic | $p$-value |
|---|---|---|---|---|
| Intercept | −10.6513 | 0.3612 | −29.5 | $< 0.0001$ |
| balance | 0.0055 | 0.0002 | 24.9 | $< 0.0001$ |

A one-unit increase in balance increases the log odds of default by 0.0055. The $z$-statistic tests $H_0$: no relationship between $X$ and the odds, i.e. $H_0 : \beta_1 = 0$; here $p < 0.0001$.

**Predictions.**

$$\hat{p}(1000) = \frac{e^{-10.6513+0.0055\times 1000}}{1 + e^{-10.6513+0.0055\times 1000}} = 0.006, \qquad \hat{p}(2000) = \frac{e^{-10.6513+0.0055\times 2000}}{1 + e^{-10.6513+0.0055\times 2000}} = 0.586.$$

**A qualitative predictor.** With student status: intercept −3.5041, student[Yes] 0.4049 ($p = 0.0004$), so $\Pr(\text{default} \mid \text{student}) = e^{-3.0992}/(1 + e^{-3.0992}) = 0.0431$ and $\Pr(\text{default} \mid \text{non-student}) = e^{-3.5041}/(1 + e^{-3.5041}) = 0.0292$.

## 6 Multiple Logistic Regression and Confounding

$$\log \frac{p(X)}{1-p(X)} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p, \qquad p(X) = \frac{e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}{1+e^{\beta_0+\beta_1 X_1+\cdots+\beta_p X_p}}.$$

| | Coefficient | Std. error | $z$-statistic | $p$-value |
|---|---|---|---|---|
| Intercept | $-10.8690$ | 0.4923 | $-22.08$ | $< 0.0001$ |
| balance | 0.0057 | 0.0002 | 24.74 | $< 0.0001$ |
| income (thousands) | 0.0030 | 0.0082 | 0.37 | 0.7115 |
| student[Yes] | $-0.6468$ | 0.2362 | $-2.74$ | 0.0062 |

Why is the student coefficient now *negative* when it was positive alone? **Confounding**: students tend to carry higher balances, so their marginal default rate is higher; but at each fixed balance, students default *less* than non-students. Multiple logistic regression teases the two apart: the marginal association mixes the direct (negative) effect with the indirect path through balance.

## 7 Extensions of Logistic Regression

- **Polynomial logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d$. Modeling the event $\texttt{wage} > 250$ with a degree-4 polynomial in age gives the fitted probability with 95% bands (wide where data are scarce).
- **Piecewise-constant logistic regression**: $\log \frac{P(x_i)}{1-P(x_i)} = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i)$ with step functions of age.
- **GAM for classification**: $\log \frac{P(X)}{1-P(X)} = \beta_0 + f_1(X_1) + \cdots + f_p(X_p)$; for the Wage data, $\beta_1\,\texttt{year} + f_2(\texttt{age}) + f_3(\texttt{education})$ with a smoothing spline (5 df) in age and a step function for education.
- **Penalized logistic regression**: minimize $-\log \mathcal{L}(\beta_0, \beta) + \lambda\|\beta\|_2^2$ (ridge) or $+\lambda\|\beta\|_1$ (lasso), generally $+\lambda\|\beta\|_q^q$; the penalty also gives finite estimates under separation.
- **Multinomial logistic regression** (below).

## 8 More Than Two Classes: Multinomial Regression

With $K$ classes (e.g. stroke, drug overdose, epileptic seizure), choose class $K$ as baseline and model, for $k = 1, \ldots, K-1$,

$$\log \frac{\Pr(Y = k \mid X = x)}{\Pr(Y = K \mid X = x)} = \beta_{k0} + \beta_{k1} x_1 + \cdots + \beta_{kp} x_p.$$

**Theorem: implied probabilities**

$$\Pr(Y = k \mid X = x) = \frac{e^{\beta_{k0}+\beta_k^\top x}}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}} \quad (k < K),$$

$$\Pr(Y = K \mid X = x) = \frac{1}{1+\sum_{l=1}^{K-1} e^{\beta_{l0}+\beta_l^\top x}},$$

and the log odds between any two classes $k, k'$ is linear: $\log \frac{\Pr(Y=k|x)}{\Pr(Y=k'|x)} = (\beta_{k0} - \beta_{k'0}) + (\beta_k - \beta_{k'})^\top x$.

**Proof.** Write $\pi_k = \Pr(Y = k \mid x) = \pi_K e^{\eta_k}$ for $k < K$; the probabilities sum to 1, so $\pi_K(1+\sum_l e^{\eta_l}) = 1$. The difference of two log odds against $K$ gives the pairwise log odds. ■

The choice of baseline does not affect the fitted probabilities, only the parametrization. A symmetric version (softmax) uses $\Pr(Y = k \mid x) = e^{\eta_k} / \sum_{l=1}^{K} e^{\eta_l}$ with one set of coefficients per class (identifiable only up to a common shift).

## Summary

- Discriminative methods model $\Pr(Y \mid X)$ directly; generative methods model $\Pr(X \mid Y)\Pr(Y)$.
- Logistic regression models the log odds linearly (a Bernoulli GLM with logit link) and is fitted by maximum likelihood (IRLS).
- Coefficients are changes in log odds; multiple regression reveals confounding (student effect reverses sign given balance).
- Polynomial, step-function, GAM and penalized versions extend it; multinomial regression handles $K > 2$ classes with linear pairwise log odds.